Answers and talk notes

Apply Area Concepts to Solve Problems

What your student is learning

Tonight we find the area of shapes made from other shapes — and decide for each piece whether to ADD it or SUBTRACT it.

Answers

Ask your student to explain before you look.

1. **A.** Subtracted. Subtracted: a cut-out piece is taken away from the total.
2. **C.** 41 square units. 41 square units: 7 × 5 = 35, 3 × 2 = 6, and 35 + 6 = 41.
3. **B.** 48 square units. 48 square units: 10 × 6 = 60, 4 × 3 = 12, and 60 − 12 = 48.
4. **D.** 52 square units. 52 square units: 8 × 5 = 40, ½ × 8 × 3 = 12, and 40 + 12 = 52.

Talk about it

Which piece did you add, which did you subtract, and how did you decide?

Home mission

Look at a wall at home. Pretend it is 10 feet wide and 8 feet tall with one window that is 3 feet by 3 feet. How much wall needs paint? Which piece do you subtract?

When they are stuck, ask

- Is this piece part of the figure, or missing from it?
- What simple shapes can you break this into?
- Did every piece get counted exactly once?

Try not to say

- Add all the pieces, that is what composite means.
- Find the big rectangle and stop there.
- Overlapping pieces are fine, just add them.